ROCKS AND MINERALS WORD SEARCH

Activity
Find the following words in the grid. They can go forward, backwards, up, down or diagonally.

Q	L	A	Y	E	R	Q	G	A	J	P	B	G	T	R	C	C	Q	R	Y	F	K
N	U	P	L	W	G	F	R	S	F	C	V	S	H	E	C	A	W	M	I	U	C
F	I	A	U	Y	B	E	A	D	N	L	I	Y	N	P	R	M	B	S	C	V	O
A	H	A	R	X	H	L	N	Q	X	H	O	O	F	S	Y	C	B	E	I	S	R
S	C	T	R	T	P	D	I	G	C	L	I	Z	S	T	S	O	A	D	K	B	O
I	I	I	W	G	Z	S	T	S	I	T	G	A	U	Q	T	N	S	I	E	N	N
T	N	V	M	F	S	P	E	T	I	P	N	C	U	E	A	G	A	M	N	A	E
Y	O	P	E	E	T	A	E	S	Y	D	E	H	C	G	L	L	L	E	I	I	N
P	T	R	R	L	A	R	O	T	S	F	O	E	H	N	U	O	T	N	V	D	O
U	U	O	B	S	B	P	A	T	I	B	U	M	Z	E	E	M	K	T	I	I	T
M	L	L	L	B	M	R	O	D	Y	C	S	I	M	I	N	E	R	A	L	S	S
I	P	A	L	O	A	N	A	S	S	T	L	C	B	S	D	R	O	R	O	B	E
C	T	T	C	G	E	G	U	M	T	T	M	A	W	S	M	A	G	Y	X	O	M
E	E	W	C	I	N	A	C	L	O	V	Y	L	C	P	C	T	G	S	C	B	I
C	E	M	E	T	A	M	O	R	P	H	I	C	B	Z	H	E	S	I	K	T	L

basalt
calcite
chemical
composition
conglomerate
crystal
feldspar
gabbro
gneiss
grain
granite
igneous
layer
limestone
marble
metamorphic
mica
mineral
obsidian
olivine
ores
plutonic
pumice
quartz
rhyolite
rock
sandstone
schist
sedimentary
shale
slate
volcanic

ROCKS AND MINERALS WORD SEARCH

ANSWERS

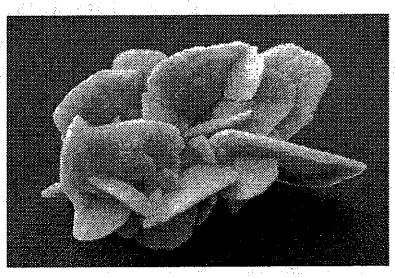
Q	L	A	Y	E	R		G						T	R							K
N	U		L			F	R					S	H		C						C
F	I	A				E	A				I	Y	N		R			S			O
A	H	A	R			L	N			H	O	O			Y	C	B	E			R
S	C		R	T		D	I		C	L	I		S		S	O	A	D			
	I	I		G	Z	S	T	S	I	T	G	A			T	N	S	I	E	N	
	N		M		S	P	E	T	I		N	C			A	G	A	M	N	A	E
	O		E	E		A	E	S		D	E	H		G	L	L	L	E	I	I	N
P	T	R	R	L		R	O	T	S		O	E		N		O	T	N	V	D	O
U	U	O	B	S	B	P		T	I		U	M		E		M		T	I	I	T
M	L		L	B	M	R	O			C	S	I	M	I	N	E	R	A	L	S	S
I	P	A		O	A	N	A				L	C		S		R		R	O	B	E
C	T		C		E	G		M				A		S		A		Y		O	M
E			C	I	N	A	C	L	O	V		L	C			T					I
		M	E	T	A	M	O	R	P	H	I	C				E					L

MINERALS AND ROCKS

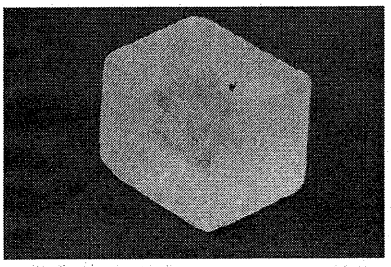
Three Level Guide

Minerals have been studied and mined throughout history. People mined **minerals** from the ground to use as they were found, particularly gemstones, and later mined **ores** to extract metals such as Gold, Copper or Iron. In fact ages of human civilisation have been named after the mineral ores used – the Bronze Age and the Iron Age for example.

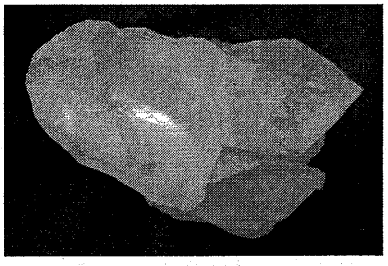
A mineral is a naturally occurring solid that has a regular **crystal structure** and **chemical make-up**. Some minerals are made up of only one kind of atom. These minerals are called 'native' minerals and include gold and silver. Other minerals are made up of several or many different types of atoms. The mineral quartz (SiO_2) is made up of molecules consisting of one atom of silicon (Si) and two atoms of oxygen (O). Quartz forms regular 18-sided crystals. Quartz is the most common mineral in the Earth's crust.



Gypsum



Calcite



Quartz

There are over 3000 different minerals and many of these are common in **rocks**, however there are only about eight very common rock-forming minerals. Examples of common rock forming minerals include quartz, feldspar, mica, olivine and calcite. Some rocks are made up of several minerals (e.g. Granite) and some rocks are made of only one mineral (e.g. limestone).

The appearance of rocks depends mainly on the type of minerals that make it up and the way the rock has been formed. Some rocks are made up of other broken down rocks that have been 'cemented' together and some are formed from interlocking crystals. The size of the mineral crystals or grains in rocks varies from very large (crystals about 5mm in granite) to microscopic (as in shale) making them very hard to see and identify. Mineral crystals are easiest to identify in rocks formed from slowly cooling **molten rock** deep under ground.

Here is a list of common rocks found in the Earth's crust and the main minerals that make them up:

- Sandstone – quartz and feldspar
- Limestone – calcite
- Granite – feldspar, quartz and mica
- Basalt – feldspar, pyroxene and olivine
- Rhyolite – quartz and feldspar

Three Level Guide

Level 1 *Reading on the lines*

Tick those statements that appear in the text. They may be worded differently but have the same meaning. Be ready to give evidence from the text.

- There are more than 300 known minerals.
- Gold, copper and iron are found in ores.
- Quartz consists of two different types of atoms.
- Gems are minerals.
- The mineral olivine is not found in rocks.
- Rocks are formed from minerals.
- Granite is a mineral.
- Minerals have consistent chemical composition.

Level 2 *Reading between the lines*

Tick those statements that you think are true from what the text suggests. Find evidence in the text to support your answer.

- Rhyolite and sandstone are made up of the same minerals but look quite different.
- Granite forms deep under ground from molten rock that cools.
- All rocks are formed in the same manner.
- In the Iron Age people learned how to extract iron from its ores.
- Quartz, feldspar and mica make up a large percentage of the Earth's crust.
- All rocks are minerals but not all minerals are rocks.

Level 3 Reading beyond the lines

Tick those statements you agree with using what the text says and what you know. Be prepared to justify your response.

- Sedimentary rocks are made up of broken pieces of other rocks.
- Rocks are classified based on how they were formed.
- Gemstones are the only valuable mineral.
- There are thousands of different types of rocks.
- The Earth is the only place in the universe that minerals can be found.
- The study of minerals is an important area of science.

WEATHERING, EROSION AND LANDFORMS

Three level guide

Landforms on the Earth's surface are changed by the processes of **weathering**, **erosion**, and **deposition**. Rocks of all types are constantly breaking down due to weathering. Weathering produces fine particles that form sediment. Sediments are blown by the wind and carried by rivers to other places in a process called erosion. Sediments are deposited and accumulate in other places and may eventually be buried and reformed into sedimentary rock millions of years later. These rocks, when eventually exposed at the surface of the Earth are again subjected to weathering.

Weathering is the gradual breakdown of rocks into sediments. The weathering of rocks is an important process that forms the soils that support plant life on Earth. Rocks all over the surface of the Earth are subjected to mechanical and chemical attack. Mechanical weathering involves the physical breakdown of rocks into smaller pieces due to physical forces exerted by temperature changes, ice, water and wind. In places with a high daily temperature variation, rocks expand during the day and contract at night. This action repeated every day eventually causes layers of the rock to peel off (this process is called onion skin weathering). During the day water can seep into cracks in rocks and if it gets cold enough overnight the water can freeze and expand, breaking the rock in a process called ice wedging. The action of water can also break rocks down, particularly when the water is moving fast such as in rivers, waterfalls and crashing ocean waves. Wind can also weather rocks, particularly when it carries sand grains and blasts them against rock surfaces. Chemical weathering involves the action of water and dissolved chemicals dissolving or reacting with rock minerals.

Erosion is the movement of sediments by water, ice, wind or gravity. Water moving downhill carries sediments and rock particles with it. Fast flowing water can carry bigger particles than slow flowing water, so fast rivers usually have rocky bottoms and slow flowing rivers have sandy or muddy bottoms. Glaciers cause great weathering but also erode sediments that are pushed downhill ahead of them. Wind blows fine sediments away, especially in desert landscapes where there is little plant growth to hold the soil together. Rocks fall off cliffs and roll down slopes under the influence of gravity and are then often moved further by the action of water.

Deposition occurs in places where moving water or wind slows down and cannot carry the sediments any further. Usually the heavier particles are dropped first, and the finer particles dropped later. Deposition of sediments forms river deltas, sand and gravel bars in lakes and sand dunes. Sediments carried by glaciers are pushed by the ice into large piles called moraine.

Landforms can be formed by weathering, erosion, deposition or a combination of factors. Steep 'v-shaped' river valleys are caused by weathering and erosion, as are the broad 'u-shaped' valleys formed by glaciers. Flat flood plains and river deltas are formed by the deposition of sediments carried by rivers. Mountains are gradually worn away by weathering and erosion. Beaches are formed by the deposition of sand carried by the action of currents and waves. The surface of the Earth is constantly changing.

Three Level Guide

Level 1 *Reading on the lines*

Tick those statements that appear in the text. They may be worded differently but have the same meaning. Be ready to give evidence from the text.

- Erosion is the breaking down of rocks to form sediments.
- Weathering affects all types of rocks.
- Mechanical weathering is caused by ice, water and wind.
- Fast flowing rivers have sandy bottoms.
- Glaciers cause weathering and erosion.
- Deposition of sediments forms glaciers.
- Rivers form 'v-shaped' valleys.
- Flood plains are caused by the deposition of sediments carried by rivers.

Level 2 *Reading between the lines*

Tick those statements that you think are true from what the text suggests. Find evidence in the text to support your answer.

- Onion skin weathering is a form of mechanical weathering.
- The faster water moves, the more weathering it can cause.
- Eventually there will be no rocks left because they will all be broken down by weathering.
- Weathering, erosion, deposition and rock formation make a cycle.
- Canyons are the result of weathering and erosion.

Level 3 Reading beyond the lines

Tick those statements you agree with using what the text says and what you know. Be prepared to justify your response.

- The amount of erosion can be controlled by people to protect the land.
- Mountains are built by weathering and erosion.
- The place where sedimentary rocks were formed can be inferred from the size of the particles that make them up.
- Australia is a very old continent, that is why it is so flat.